

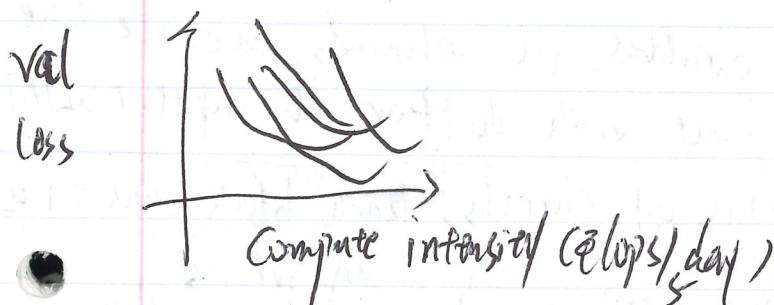
Lecture #5: GPU Architecture and performance Analysis

Make CPU and GPUs less magic

⇒ understand when GPUs get slow

⇒ understand how to make fast algorithms

Part #1: GPU in depth - how they work and important parts



How to save compute? Early on - Dennard Scaling

parallel scaling continues

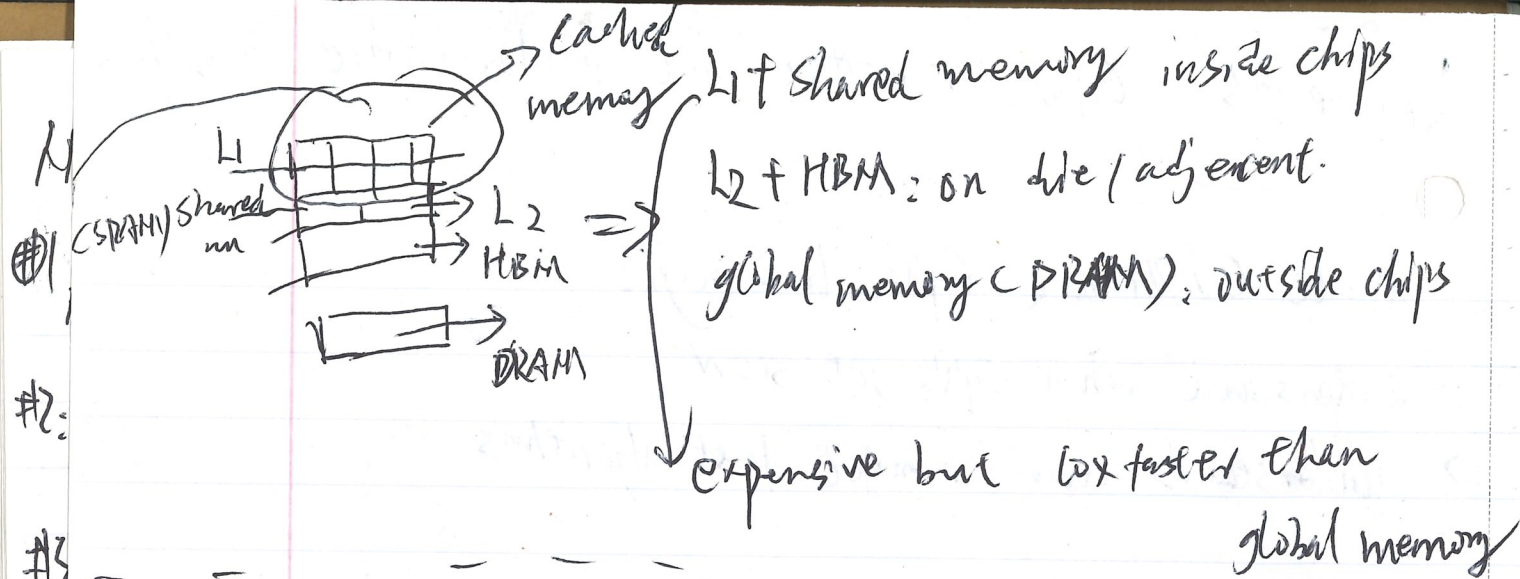
CPU: optimize a few, fast threads; opt latency (each thread finishes quickly)

GPU: optimize for many threads; optimize for throughput (total processed data)

SM: streaming multiprocessor

includes many SP (streaming processor)

↓
The closer the memory to the SM, the faster it is.
L1 and shared memory is inside the SM. L2 cache is on die, and global memory are the memory chips next to GPU

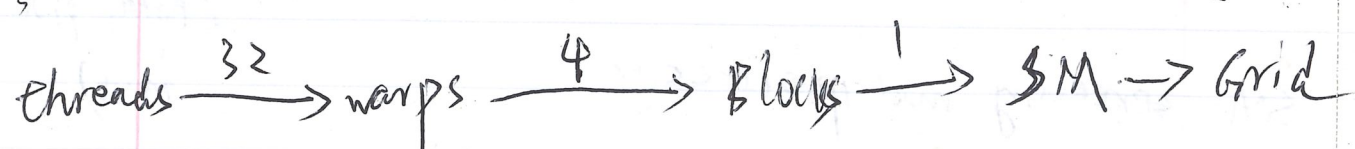
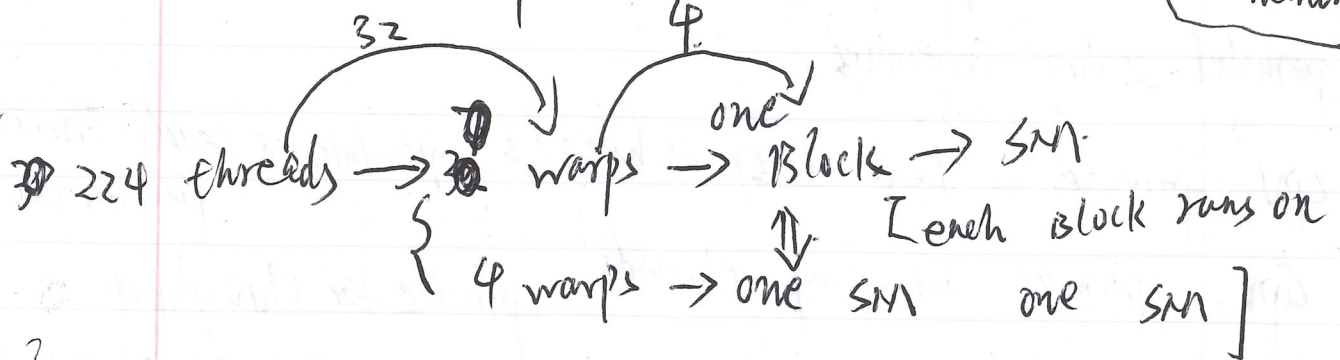


Execution model of a GPU:

Threads do the work in parallel: all threads execute the same instructions but with different inputs (SIMT)

Blocks: groups / collections of threads. Each block runs on a SM w/ its own shared memory

warp: Threads execute in a warp of 32 consecutively numbered threads each



each thread can access its own register and shared memory within the block.

~~and~~